

phase_05_GAP_AUDIT

Phase 5 Gap Audit — Advanced & Exotic Device Ecosystem

Field	Value
Auditor	Engineering auditor (code-grounded, anti-bluff)
Date	2026-06-01
Method	Direct inspection of pkg/, internal/, cmd/, api/ against PHASE_5_ROADMAP.md deliverables
Honest completion	~3% complete

3-Axis Package Status (Refreshed 2026-06-04, HXC-939)

Why this section exists: “exists” ≠ “used”. wired is **measured** via `go list -deps ./cmd/... | grep -Fx <module-path>/<pkg> (module = github.com/HelixDevelopment/helix_cluster)`, not assumed. **“Completed (registry) ≠ wired”** — Completed only means source+tests exist; it does NOT prove a shipped binary reaches it. Phase 5 is almost entirely **unimplemented**, so most rows are implemented=NO; the table makes that mechanical rather than narrative.

Package	implemented	wired (reachable from cmd/)	tested
pkg/device	NO	n/a	n/a
pkg/handheld	NO	n/a	n/a
pkg/fpga	NO	n/a	n/a
pkg/riscv	NO	n/a	n/a
pkg/cloudspot	NO	n/a	n/a
pkg/inference	NO	n/a	n/a

Package	implemented	wired (reachable from cmd/)	tested
pkg/quantum	NO	n/a	n/a
internal/ handheld / sbc / fpga / enterprise / iot / exotic	NO	n/a	n/a
pkg/ scheduler (would be <i>extended</i> for tier/power)	yes (base only)	yes	yes
pkg/ resources (would be <i>extended</i> for NPU/FPGA/ TFLOPS)	yes (base only)	yes	yes
pkg/ discovery (would be <i>extended</i> for 15-tier/trust)	yes (base only)	yes	yes
pkg/ wireguard (would be <i>extended</i> for spot teardown)	yes (base only)	yes	yes

Note: the four “extended” base packages are implemented+wired+tested for their *Phase 2–4* scope but carry **none** of the Phase 5 tier/trust/power/NPU behaviour, so they are not Phase 5 deliverables. No Phase 5 package is orphaned because no Phase 5 package exists yet.

One-line summary: Phase 5 is effectively unstated in code — none of the 7 planned pkg/ packages, none of the 6 planned internal/ packages, and none of the planned proto fields (tier, trust_level, compute_class, arch, npu_tops, power_budget) exist; the only adjacent assets are generic Phase 2–4/8C foundations (scheduler, resources, wireguard, llm manager) that were NOT extended for device-tier/trust/power-aware behaviour. Recent “Wave1–5 / Phase 8C” hardening (RBAC scopes, pkg/fiber, cost-aware GPU placement, PQ-crypto/gpuattest/TEE attestation in the security/ submodule) is real but belongs to other tracks and does not satisfy any Phase 5 deliverable.

Deliverable Status Table

The roadmap's concrete deliverables are the new `pkg/` (§4.1), new `internal/` (§4.2), extended packages (§4.3), and proto extensions (§5.3).

Deliverable	Status	Evidence (file:line)
<code>pkg/device</code> — capability descriptor + auto-probe (P0 #1)	MISSING	<code>dir</code> absent (<code>ls pkg/</code> — no <code>device</code>)
<code>pkg/handheld</code> — Steam Deck/x86 handheld agent, Vulkan/battery/thermal (P0 #2)	MISSING	<code>dir</code> absent
<code>pkg/fpga</code> — hard-SoC/soft-core/DPU backends (P1 #6)	MISSING	<code>dir</code> absent
<code>pkg/riscv</code> — riscv64 cross-compile helpers, RVV probe (P0 #5)	MISSING	<code>dir</code> absent
<code>pkg/cloudspot</code> — WireGuard join/leave + preemption handler (P1 #8)	MISSING	<code>dir</code> absent; <code>pkg/wireguard/mesh.go</code> has generic join/leave only
<code>pkg/inference</code> — Groq/Cerebras/Jetson/llama.cpp provider-agnostic backend (P1 #10)	MISSING	<code>internal/llm/manager.go:142</code> (Inference)

Deliverable	Status	Evidence (file:line)
pkg/quantum — Qiskit Runtime plugin (P2 #12)	MISSING / DEFERRED	dir absent
internal/handheld	MISSING	dir absent
internal/sbc — RK3588 NPU / Jetson TensorRT / Turing RK1	MISSING	dir absent
internal/fpga	MISSING	dir absent
internal/enterprise — EPYC/Ampere auto-provision, Coreboot trust	MISSING	dir absent
internal/iot — OpenWrt/NAS/webOS agents	MISSING	dir absent
internal/exotic — Groq/Cerebras clients, quantum, neuromorphic	MISSING	dir absent
pkg/scheduler extension — tier-aware matchmaking + power-aware T9 policy (P0 #4)	MISSING	pkg/scheduler/types.go (no Tier/Trust/PowerBudget/Arch)
pkg/resources extension — NPU TOPS / FPGA LE / GPU TFLOPS / QPU probing	MISSING	pkg/resources/types.go:24 (GPUInfo has no TFLOPS); no npu/tops/fpga/tflops fields
pkg/discovery extension — 15-tier registry + trust-aware routing	MISSING	pkg/discovery/selection.go, etcd_backend.go (no tier/trust)
pkg/wireguard extension — spot dynamic join/leave + preemption-safe teardown	PARTIAL	pkg/wireguard/mesh.go, manager.go
proto: node.proto += tier,trust_level,compute_class,arch	MISSING	api/v1/node.proto:20-45 (only node_id,region,status,reason)
proto: scheduler.proto += tier constraints, power_budget	MISSING	api/v1/scheduler.proto (grep tier/power → none)

Deliverable	Status	Evidence (file:line)
proto: resources.proto += npu_tops,fpga_logic_elements,tflops_gpu	MISSING	api/v1/resources.proto file does not exist
Steam Deck Flatpak agent (5a)	MISSING	no cmd/ or packaging artifact
Jetson TensorRT backend (5a)	MISSING / DEFERRED	—
riscv64 CI build artifact (5b)	MISSING	no riscv pipeline in scripts//CI verified
FPGA DPU / Groq / Cerebras / Cloud-spot exit gates	MISSING	—

Tally: 0 DONE, 1 PARTIAL (pkg/wireguard generic mesh), ~22 MISSING.
Hence ~3% (sole partial credit = reusable wireguard/scheduler/resources/discovery scaffolding that a Phase 5 build would extend).

TOP IMPLEMENTABLE GAPS (Go, no new infra)

Prioritized, concrete, mutation-pairable. These need no hardware and unblock the rest of Phase 5.

1. pkg/device — universal capability descriptor + arch auto-probe (P0, unblocks everything)

Target: pkg/device/ **Spec:** Define Descriptor{ Tier, TrustLevel, ComputeClass, Arch, CPUCores, MemMB, GPUs, NPUTops, FPGALogicElements, GPUTFLOPS, PowerBudgetW } with enums for the 15 tiers (T1–T15) and trust levels. Implement Probe(ctx) (Descriptor, error) that derives Arch from runtime.GOARCH, CPU/mem from existing pkg/resources readers, and reads optional override labels (env/config) for fields not auto-detectable (NPU/FPGA/TFLOPS). Provide AssignTier(Descriptor) Tier implementing the §50 taxonomy decision tree (e.g. riscv64→T10, x86+open-firmware→T1, GPU+battery present→T9). Pure Go, deterministic. **Acceptance test (mutation-pairable):** table-driven fixtures mapping synthetic Descriptors → expected tier. *Mutation:* swap the riscv64→T10 branch to →T2; the riscv fixture assertion must fail. Verifies tier logic, not just non-panic.

2. proto + codegen: device/tier fields (P0, blocks scheduler+discovery wiring)

Target: `api/v1/node.proto`, new `api/v1/resources.proto`, `api/v1/scheduler.proto` **Spec:** Add `tier`, `trust_level`, `compute_class`, `arch` to `Node`; create `resources.proto` with `npu_tops`, `fpga_logic_elements`, `tflops_gpu`; add `power_budget` + repeated tier constraints to scheduler request. Regenerate via existing buf pipeline. Add round-trip mapping helpers `device.Descriptor <=> pb.Node`. **Acceptance test:** marshal a `Descriptor`→`proto`→`Descriptor`, assert field equality incl. tier enum. *Mutation:* drop the `arch` field copy in the mapper → round-trip assertion fails.

3. pkg/scheduler tier-aware + power-aware plugin (P0 #4)

Target: `pkg/scheduler/tier_power.go` **Spec:** New Plugin (same Name/Filter/Score/Bind contract as `cost_gpu.go`) that (a) Filters out nodes whose Tier/TrustLevel violate a job's RequiredTier/MinTrust, and (b) for T9 HANDHELD nodes, Filters when `power_budget remaining < job estimate` or node reports gaming-active/on-battery label. Reuse the existing plugin registration in `plugins.go`. **Acceptance test:** feed a job requiring T1 against a mixed node set; assert only T1 nodes survive Filter, and a charging-T9 node is admitted while a battery-only T9 is rejected. *Mutation:* invert the trust comparison (`>=><`) → an untrusted node leaks through and the assertion fails.

4. pkg/discovery 15-tier registry + trust-aware selection (P0)

Target: `pkg/discovery/tier_registry.go` **Spec:** Extend the etcd-backed registry to index nodes by tier/trust and add `SelectByTier(tier, minTrust)` returning only matching entries. Backward-compatible with existing `selection.go`. **Acceptance test:** register synthetic nodes across T1/T9/T10, query `SelectByTier(T10, ...)`, assert exact set. *Mutation:* make `SelectByTier` ignore the trust filter → a below-trust node appears and assertion fails. (Run against the existing etcd integration harness for real-behavior credit.)

5. pkg/cloudspot preemption handler (logic only, no AWS) (P1 #8)

Target: `pkg/cloudspot/` **Spec:** PreemptionWatcher interface + a pluggable SignalSource (file/HTTP-poller injectable for test). On signal, fire ordered hooks: Drain → Checkpoint → Leave(mesh). Provide a FakeSignalSource for tests; real AWS/GCP metadata poller is a thin adapter (deferred to infra). Wire Leave to `pkg/wireguard` mesh teardown. **Acceptance test:** inject a fake preemption signal; assert hooks run in order and Leave is called exactly once within the 2-min budget (simulated clock). *Mutation:* reorder so Leave precedes Checkpoint → ordering assertion fails.

6. pkg/inference provider-agnostic interface + local backend (P1 #10, partial)

Target: `pkg/inference/` **Spec:** Define `Backend{ Name(); Generate(ctx, Req) (Resp, error) }` and a Router that selects a backend by capability/tier. Implement a `LocalBackend` wrapping the existing `internal/llm` manager so the

interface is exercised end-to-end today. Groq/Cerebras/Jetson are adapter structs (HTTP/SDK) marked DEFERRED until credentials/hardware exist, but the interface + routing + local path are fully testable now. **Acceptance test:** register a fake + local backend; route a request and assert the chosen backend matches the tier policy and response propagates. *Mutation:* make Router always pick index 0 → the tier-routing assertion fails.

DEFERRED (external infra / hardware / Zig / GPU-kernel — cannot be honestly completed in-repo)

Deliverable	Reason deferred
Steam Deck Flatpak agent gaming-interference gate	Requires real Steam Deck; exit gate demands real-device FPS/battery benchmark (CLAUDE-1 forbids mock-only).
Jetson TensorRT ≥ 60 TOPS backend	Requires Jetson Orin L4T hardware + TensorRT runtime.
FPGA hard-core (DE10-Nano/Zynq/KV260 DPU) + soft-core (Colorlight/VexRiscv)	Requires physical FPGA boards + Vitis AI / Yosys-LiteX toolchains.
riscv64 native CI build artifact	Implementable but needs riscv64 toolchain/runner in CI infra.
EPYC / Ampere Altra auto-provisioning gate	Requires real bare-metal server hardware for <30-min onboarding evidence.
Cloud spot hybrid (5 on-prem + 5 spot) integration gate	Requires live AWS/GCP spot accounts.
Groq LPU <100ms TTFT / Cerebras CS-3	Requires GroqCloud / Cerebras cloud API credentials.
pkg/quantum Qiskit Runtime	Requires IBM Quantum account; research-only T14.
webOS smart TV agent	Requires webOS device + JS bridge runtime.
gVisor/Kata per-tier sandbox hardening	Requires gVisor/Kata runtime on real cluster nodes.

Deferred count: 10 — all blocked on external hardware, cloud accounts, or device-specific toolchains; none are pure-Go in-repo work. The 6 implementable gaps above (pkg/device, proto fields, tier/power scheduler plugin, tier-aware discovery, pkg/cloudspot logic, pkg/inference interface+local backend) constitute the honest, no-new-infra path to move Phase 5 off ~3%.